

# Carré du champ Flow Matching

Better Quality-Generalisation Tradeoff in Generative Models

**Generalisation**

*VS*

**Memorisation**

# Flow Matching: Review

Learn a velocity field

$$u_t(x) : [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}^d, \quad \xrightarrow{\text{generates}} \quad p_t(x)$$

Satisfying the continuity equation

$$\frac{\partial}{\partial t} p_t(x) = -\nabla \cdot (u_t(x) p_t(x)), \quad p_0 = \mu(x), \quad p_1(x) = \nu(x)$$

Consider

$$\mu(x) = \mathcal{N}(x; 0, I)$$

Flow path of particles

$$X_t := \psi_t(X_0) \sim p_t$$

Characteristic ODE

$$\frac{d}{dt} \psi_t(x) = u_t(\psi_t(x)), \quad \psi_0(x) = x$$

# Flow Matching: Review

Final particle position

$$X_1 \sim \nu \xrightarrow{\text{specifies}} \psi_t(X|X_1)$$

Choose an affine flow

$$\psi_t(X|X_1) = tX_1 + \sigma_t X, \quad \sigma_t := (1 - t) + t\sigma_{min}$$

Probability path

$$p_t(x|x_1) = \mathcal{N}(x \mid tx_1, \sigma_t^2 I)$$

# The Risk of Memorisation

Probability path

$$p_t(x|x_1) = \mathcal{N}(x \mid tx_1, \sigma_t^2 I)$$

Marginalising over  $\nu(x_1)$  as  $t \rightarrow 1$ ,  $\sigma_{\min} \downarrow 0$

$$\nu \approx p_1(x) = \int_{\mathbb{R}^d} \mathcal{N}(x \mid x_1, \sigma_{\min}^2 I) \nu(x_1) dx_1 \approx \frac{1}{N} \sum_{i=1}^N \mathcal{N}(x \mid x^{(i)}, \sigma_{\min}^2 I)$$

# The Bochner-Weitzenböck Formula

$$\frac{1}{2}\Delta(|\nabla f|^2) = \langle \nabla f, \nabla(\Delta f) \rangle + |\nabla^2(f)|^2 + \text{Ric}(\nabla f, \nabla f)$$

# The Carré du Champ Operator ( $\Gamma$ )

$$\Gamma(f, h) = \frac{1}{2}(L(fh) - fL(h) - hL(f)) = \langle \nabla f, \nabla h \rangle$$

$$\Gamma(f, f) = \frac{1}{2}(L(f^2) - 2fL(f))$$

$$\Gamma(f, f) = |\nabla f|^2$$

## The Iterated Carré du Champ Operator ( $\Gamma_2$ )

$$\Gamma_2(f, f) = \frac{1}{2} \Delta \Gamma(f, f) - \Gamma(f, \Delta f)$$



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# Estimating the Carré du Champ

Variable-bandwidth Gaussian kernel

$$w_{\epsilon}(x^{(i)}, x^{(j)}) = \exp \left( -\frac{\|x^{(i)} - x^{(j)}\|^2}{\epsilon_i \epsilon_j} \right)$$

Local transition probabilities (Markov kernel)

$$P_{ij} = \frac{w_{\epsilon}(x^{(i)}, x^{(j)})}{\sum_l w_{\epsilon}(x^{(i)}, x^{(l)})}$$

Optimal local covariance estimation

$$\hat{\Gamma}(x^{(i)}) = \mathbb{E}_{X \sim P_i} \left[ (X - m^*(x^{(i)}))(X - m^*(x^{(i)}))^T \right]$$

$$\psi_t(X \mid X_1) = tX_1 + \sigma_t X, \quad \sigma_t := (1 - t) + t\sigma_{min}$$

$$\psi_t^\Gamma(X \mid X_1) = tX_1 + \Sigma_t^\Gamma(X_1)^{\frac{1}{2}}X, \quad \Sigma_t^\Gamma(x_1) = \left[ (1 - t)I + t\hat{\Gamma}(x_1)^{1/2} \right]^2$$

$$p_t(x \mid x_1) = \mathcal{N}(x \mid tx_1, \Sigma_t^\Gamma(x_1))$$



## Generalisation

Negative Log-Likelihood (NLL) on test set

## Quality

Distance-to-Manifold (DtM), Fréchet Inception Distance (FID)

## Memorisation (Nearest-Neighbour Ratio)

$$M(y) := \frac{\|y - x^{(1)}\|}{\|y - x^{(2)}\|} < \text{cutoff}$$

$x^{(1)}, x^{(2)}$  are the 1st and 2nd nearest training neighbours to sample  $y$ .

# Experimental Setup

## 1. Spatially Heterogeneous Data

*Testing the model's response to extreme density variations.*

- **Synthetic:** The Two-Circles Dataset
- **Real-World:** Drosophila (Fruit Fly) Motion Capture (372D)

## 2. Scaling and Dimensional Limits

*Testing the boundaries of the quality-generalisation tradeoff.*

- **Curse of Dimensionality:** The Torus Dataset
- **High-Dimensional Image Domains:** CIFAR-10 and CelebA-HQ